NATIONAL TEACHERS COLLEGE

"CarbonTrack: SOAST Students Carbon Footprint Monitoring System"

Finals Project
Presented to the Faculty of the
School of Arts, Science and Technology
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In Partial Fulfillment of the
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for the Degree of
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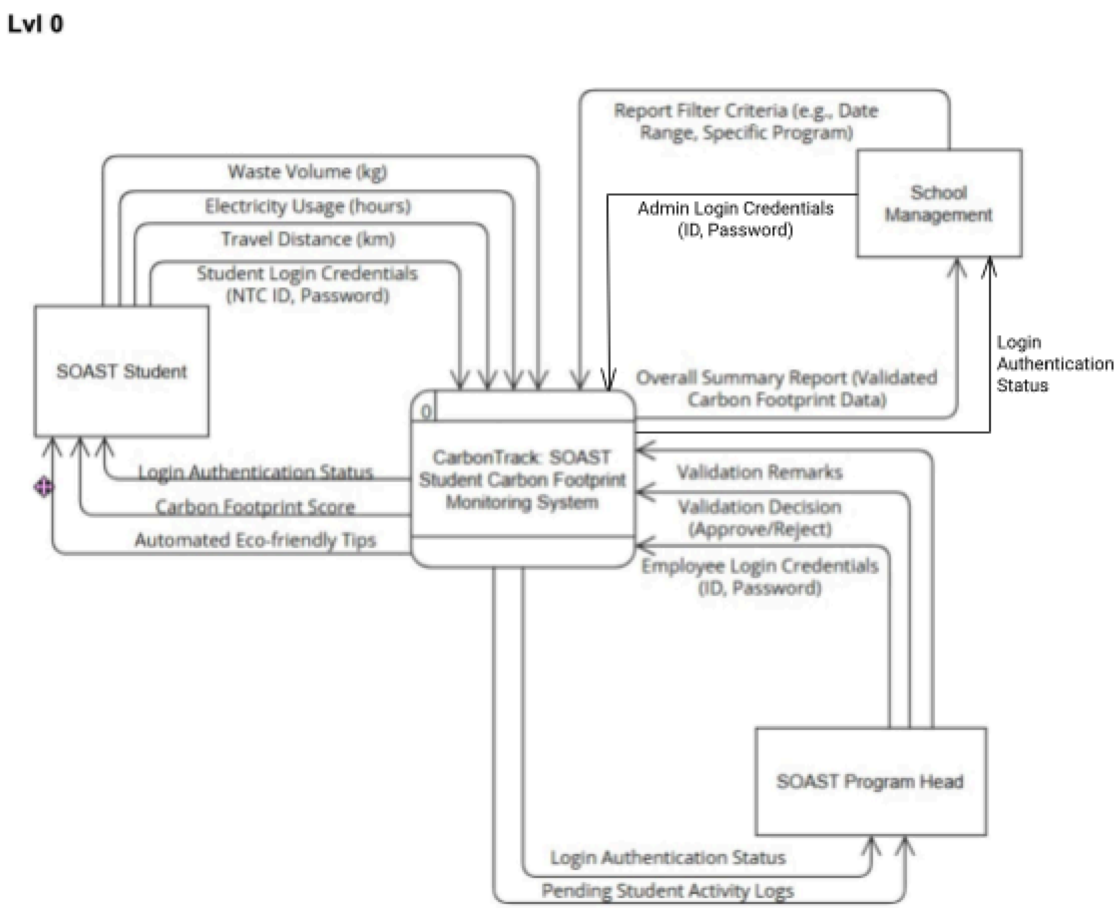
I. Process Architecture

The Process Architecture of the **Carbon Track: SOAST Students Carbon Footprint Monitoring System** is designed to establish a clear, secure, and logical flow of information. The Level-0 and Level-1 Data Flow Diagrams (DFDs) illustrate the system's core processes, representing the precise movement of data from external entities such as student inputs and program head validations through the system's internal computation, and finally into aggregated management reports.

Level 0 (Context Diagram)

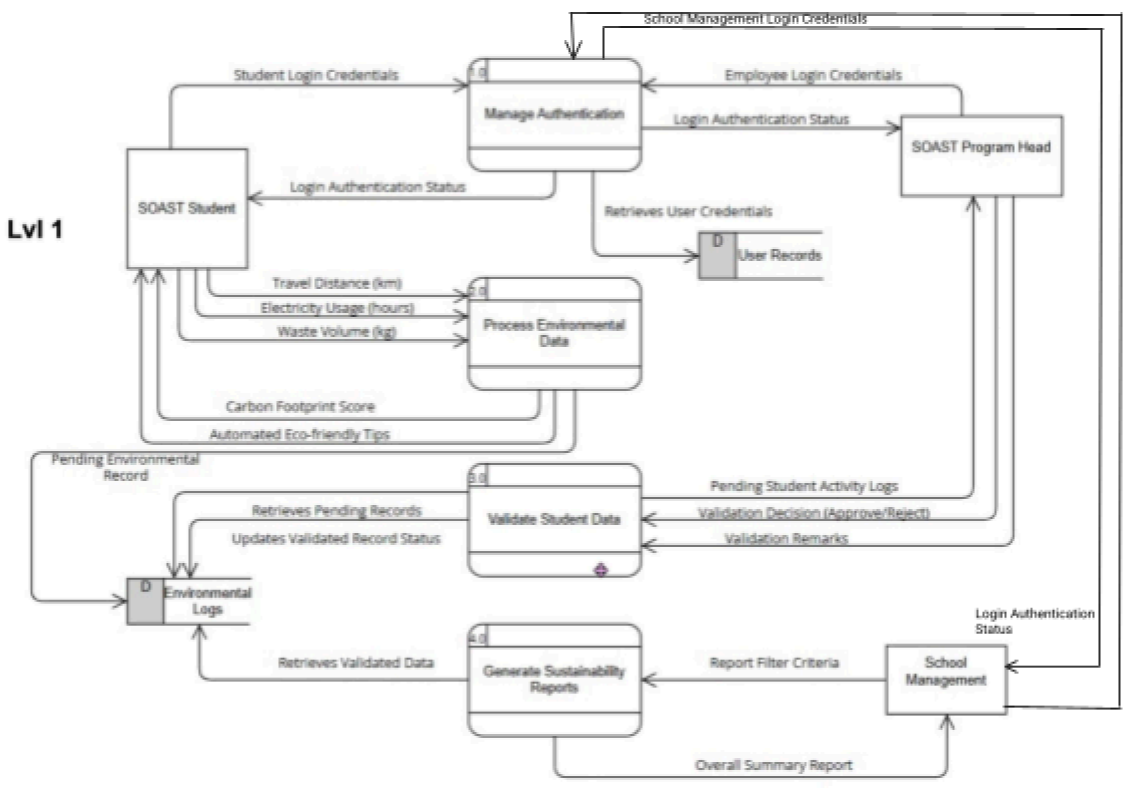


Hierarchy: CarbonTrack System Level 0 DFD
External Entities: SOAST Student, SOAST Program Head, and School Management

Data Flows:
SOAST Student sends Student Login Credentials to CarbonTrack System
SOAST Student sends Travel Distance (km) to CarbonTrack System

SOAST Student sends Device Usage (hours) to CarbonTrack System
SOAST Student sends Waste Volume (kg) to CarbonTrack System
CarbonTrack System sends Login Authentication Status to SOAST Student
CarbonTrack System sends Carbon Footprint Score to SOAST Student
CarbonTrack System sends Automated Eco-friendly Tips to SOAST Student
SOAST Program Head sends Employee Login Credentials to CarbonTrack System
SOAST Program Head sends Validation Decision (Approve/Reject) to CarbonTrack System
SOAST Program Head sends Validation Remarks to CarbonTrack System
CarbonTrack System sends Login Authentication Status to SOAST Program Head
CarbonTrack System sends Pending Student Activity Logs to SOAST Program Head
School Management sends Report Filter Criteria to CarbonTrack System
CarbonTrack System sends Overall Summary Report to School Management

Level 1 (Process Explosion)



Hierarchy: CarbonTrack System Level 1 DFD
Processes:
1.0 Manage Authentication
2.0 Process Environmental Data
3.0 Validate Student Data
4.0 Generate Sustainability Reports

External Entities: SOAST Student, SOAST Program Head, and School Management
Data Stores: D1: User Records D2: Environmental Logs
Data Flows:
(Authentication Process)
SOAST Student sends Student Login Credentials to 1.0 Manage Authentication
SOAST Program Head sends Employee Login Credentials to 1.0 Manage Authentication
1.0 Manage Authentication retrieves User Credentials from D1: User Records
1.0 Manage Authentication sends Login Authentication Status to SOAST Student
1.0 Manage Authentication sends Login Authentication Status to SOAST Program Head
1.0 Manage Authentication sends Login Authentication Status to School Management

(Data Processing Process)
SOAST Student sends Travel Distance (km) to 2.0 Process Environmental Data
SOAST Student sends Device Usage (hours) to 2.0 Process Environmental Data
SOAST Student sends Waste Volume (kg) to 2.0 Process Environmental Data
2.0 Process Environmental Data stores Pending Environmental Record to D2: Environmental Logs
2.0 Process Environmental Data sends Carbon Footprint Score to SOAST Student
2.0 Process Environmental Data sends Automated Eco-friendly Tips to SOAST Student

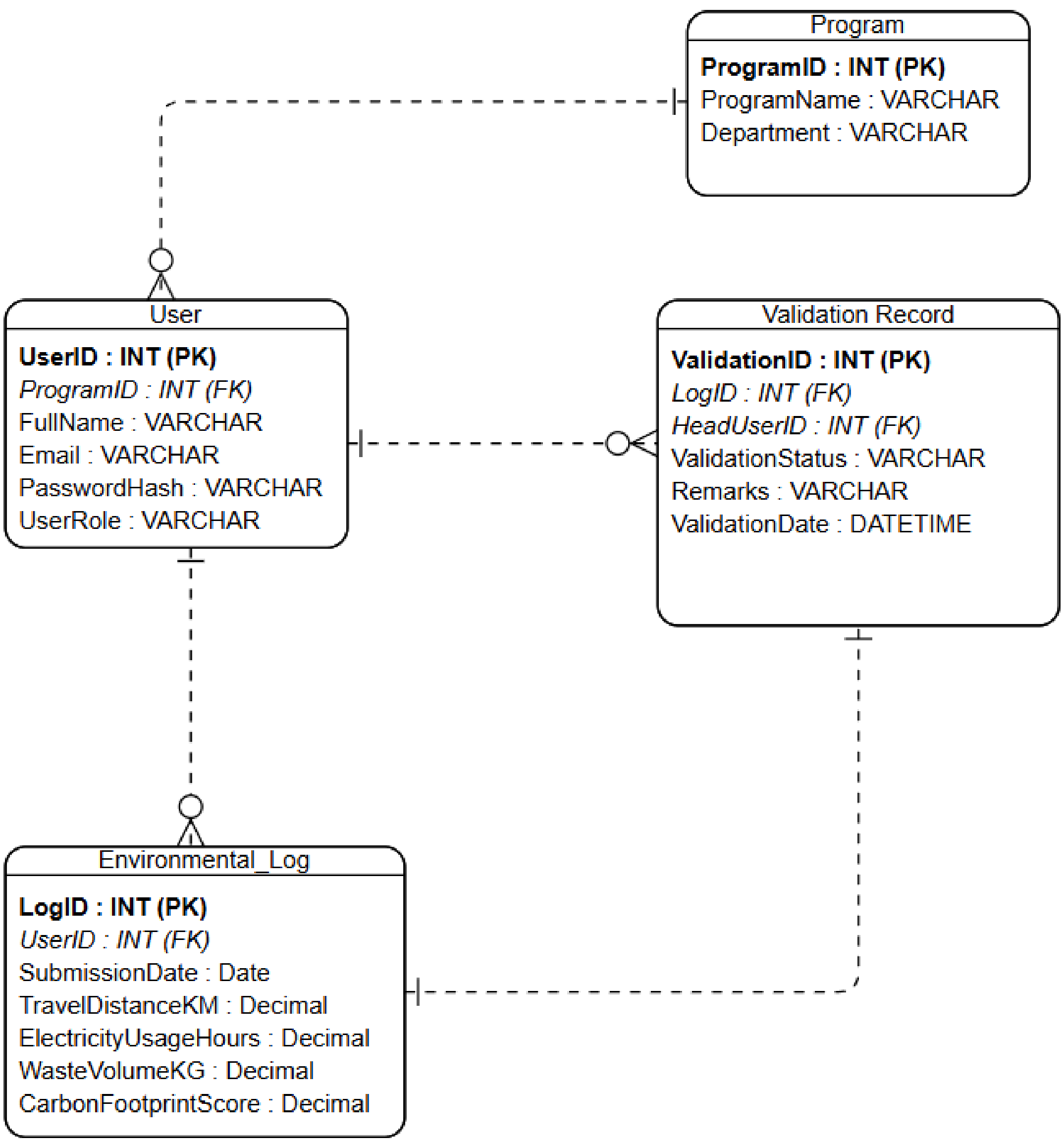
(Validation Process)
3.0 Validate Student Data retrieves Pending Records from D2: Environmental Logs
3.0 Validate Student Data sends Pending Student Activity Logs to SOAST Program Head
SOAST Program Head sends Validation Decision (Approve/Reject) to 3.0 Validate Student Data
SOAST Program Head sends Validation Remarks to 3.0 Validate Student Data
3.0 Validate Student Data updates Validated Record Status in D2: Environmental Logs

(Reporting Process)
School Management sends Report Filter Criteria to 4.0 Generate Sustainability Reports
4.0 Generate Sustainability Reports retrieves Validated Data from D2: Environmental Logs
4.0 Generate Sustainability Reports sends Overall Summary Report to School Management

II. Data Architecture

The Data Architecture of the **Carbon Track: SOAST Students Carbon Footprint Monitoring System** is designed to ensure data integrity, relational accuracy, and efficient backend storage. The logical Entity-Relationship Diagram (ERD) illustrates the foundational database structure, representing the interconnected tables, key constraints, and cardinality rules required to reliably manage user credentials, track transactional footprint logs, and store automated eco-tips.

Entity Relationship Diagram (ERD)



3rd Normal Form (3NF)

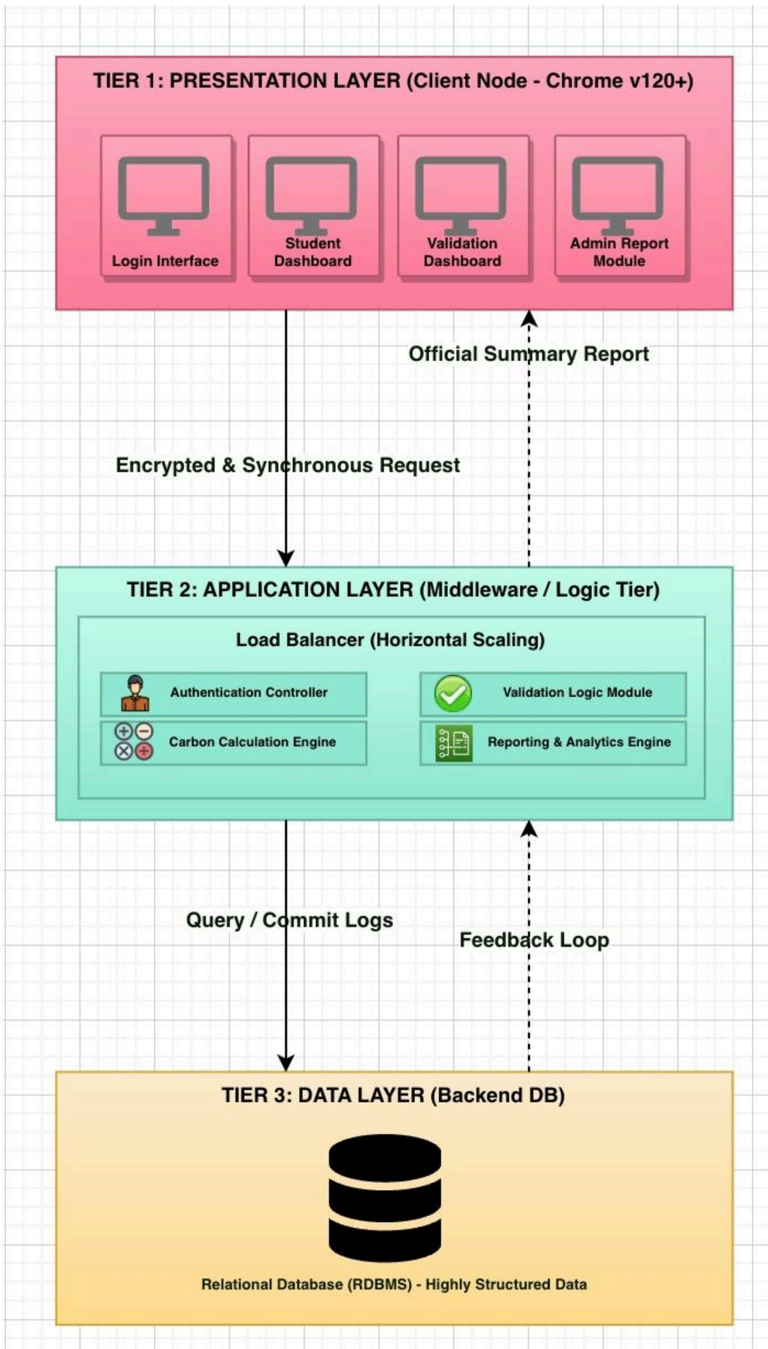
Entity	Attributes	Data Type	Key Constraints
Program	**ProgramID**	INT	**PK** (Primary Key)
	ProgramName	VARCHAR	
	Department	VARCHAR	
User	**UserID**	INT	**PK** (Primary Key)
	ProgramID	INT	**FK** (References PROGRAM)
	FullName	VARCHAR	
	Email	VARCHAR	
	PasswordHash	VARCHAR	
	UserRole	VARCHAR	*(Student / Program Head / School Management)*
Environmental Log	**LogID**	INT	**PK** (Primary Key)
	UserID	INT	**FK** (References USER)
	SubmissionDate	DATE	
	TravelDistanceKM	DECIMAL	
	ElectricityUsageHours	DECIMAL	
	WasteVolumeKG	DECIMAL	
	CarbonFootprintScore	DECIMAL	
Validation Record	**ValidationID**	INT	**PK** (Primary Key)
	LogID	INT	**FK** (References ENVIRONMENTAL_LOG)

	HeadUserID ValidationStatus *(Approved/Rejected)* Remarks ValidationDate	INT VARCHAR VARCHAR DATETIME	**FK** (References USER) *(Approved / Rejected / Pending)*

III. System Infrastructure

The System Infrastructure of the **Carbon Track: SOAST Students Carbon Footprint Monitoring System** is designed to provide a secure, scalable, and highly organized technical foundation. The 3-Tier Architecture sketch illustrates the strict separation of concerns within the system, representing the distinct operational boundaries between the client-facing Presentation layer, the centralized Application logic layer, and the secured backend Data storage layer.

3-Tier Architecture



IV. TELOS Feasibility

The TELOS Feasibility Analysis of the **Carbon Track: SOAST Students Carbon Footprint Monitoring System** is designed to provide a comprehensive and objective evaluation of the project's overall viability. This assessment illustrates the strategic readiness of the proposed solution, representing a thorough validation of its Technical capabilities, Economic efficiency, Legal compliance to data privacy, Operational workflow integration, and Scheduling constraints prior to full-scale development.

Technical Feasibility

The **CarbonTrack** system is technically feasible because the development team utilizes widely available and reliable resources such as HTML, CSS, JavaScript, and SQL for database management. The system is designed as a web-based platform that can operate through standard web browsers without requiring specialized hardware or Internet of Things (IoT) sensors. This reduces technical complexity while still allowing the implementation of essential features such as the carbon calculation engine, user logging portal, and environmental dashboard. Furthermore, the use of open-source development tools ensures that the team effectively builds, tests, and deploys the prototype within the project timeline.

Economic Feasibility

The project is economically feasible since the expected benefits outweigh the development costs. The system will be developed using free and open-source software applications, including Visual Studio Code and free web hosting platforms, eliminating the need for expensive licensing fees or proprietary software. In addition, the project does not require the procurement of advanced hardware devices because data collection will rely on manual input. By automating sustainability reporting and carbon emission calculations, the system can significantly reduce the time and effort required for manual report preparation, resulting in operational efficiency for the institution.

Legal Feasibility

The proposed system is legally feasible because it will comply with the provisions of the Philippine Data Privacy Act of 2012 (RA 10173). The system will implement privacy and security measures to protect the personal information of students, faculty, and staff. User-submitted records will remain confidential, while the environmental dashboard will display only summarized or aggregated emission data to prevent unauthorized disclosure of identities. In addition, all collected data will be used solely for sustainability monitoring and reporting purposes within the institution.

Operational Feasibility

The system is operationally feasible because it addresses the institution's current challenge of managing sustainability reports through manual and fragmented processes. The proposed web-based platform provides a centralized and user-friendly solution that allows students and faculty to conveniently log transportation and waste data using computers or mobile devices. The instant generation of environmental reports and visual dashboards can improve monitoring efficiency and encourage active participation from stakeholders. Since the system is accessible through standard web browsers, users can easily integrate it into their daily routines with minimal training requirements.

Schedule Feasibility

The project is schedule feasible because the development activities are organized within a realistic and manageable timeline. The project team has allocated sufficient time for each major phase of development, including system design, database preparation, coding, testing, and final deployment. The proposed schedule is as follows:

Phase	Target Schedule	Activities
Planning and Design	April (Weeks 1 - 2)	Finalization of system interface design and database structure
Development Phase	April (Weeks 3 - 4)	Coding of the carbon calculator, user authentication, and logging modules
Testing and Debugging	May (Week 1 - 2)	System testing, validation, and correction of identified errors
Final Deployment	May 15, 2026	Submission and presentation of the working web prototype

V. UI/UX Blueprint and Core User Journey

The UI/UX Blueprint of the **Carbon Track: SOAST Students Carbon Footprint Monitoring System** is designed to provide an intuitive, role-based, and seamless user experience. The high-fidelity mockups illustrate the core user journey, representing the complete lifecycle of data from initial entry by students to validation by program heads, and finally, aggregation by the management.

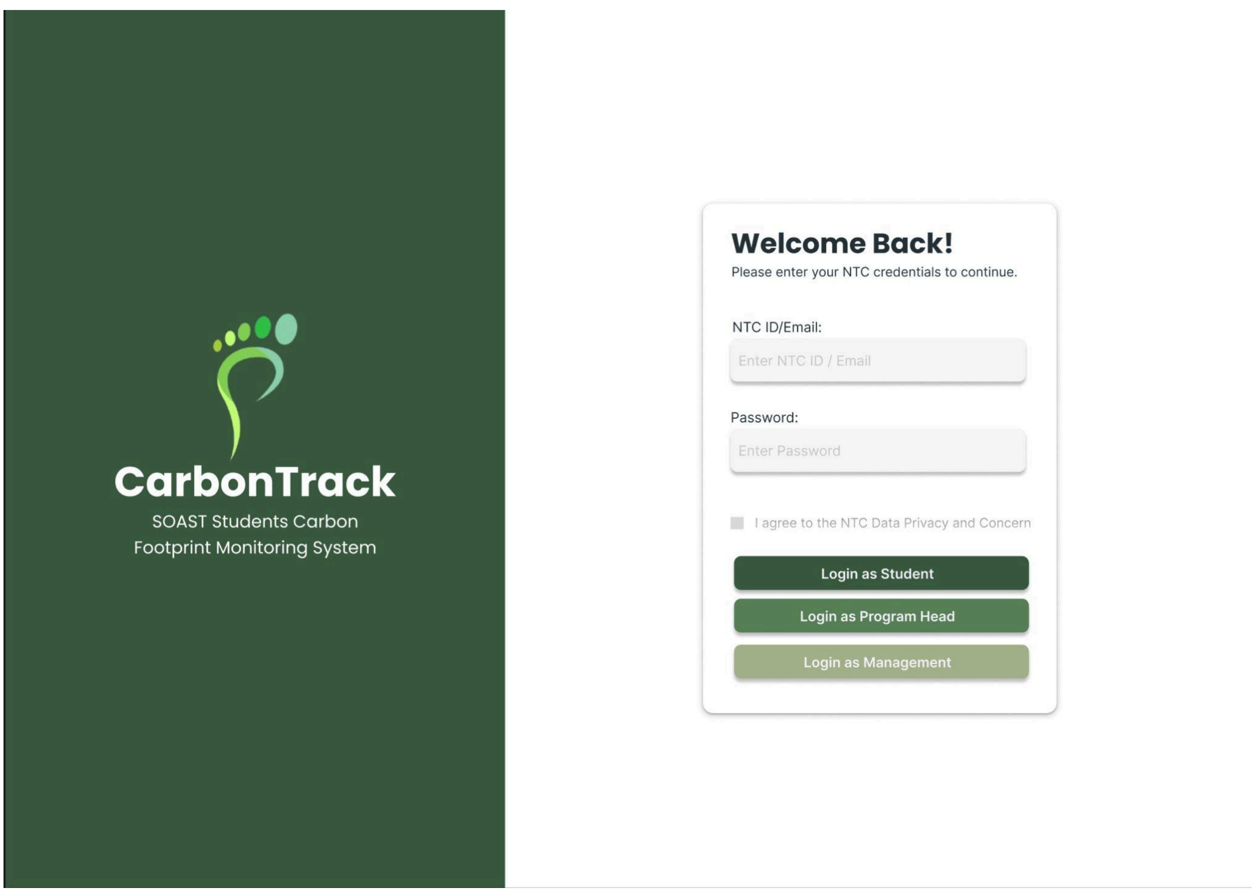
Core User Journey Mapped to High-Fidelity Mockups

The core user journey is divided into four main interface modules, each corresponding to a specific step in the system's process flow:

Module 1: The Authentication Checkpoint (Login Interface)

Purpose: Serves as the secure entry point for all system users.
Key Features:

- **Credential Verification:** Requires users to input their designated "NTC ID/Email" and "Password".
- **Compliance:** Features a mandatory checkbox stating "I agree to the NTC Data Privacy and Concern" to ensure data handling compliance before access is granted.
- **Role-Based Access Control (RBAC):** Users are directed to their specific environments using dedicated access buttons: "Login as Student," "Login as Program Head," and "Login as Management".

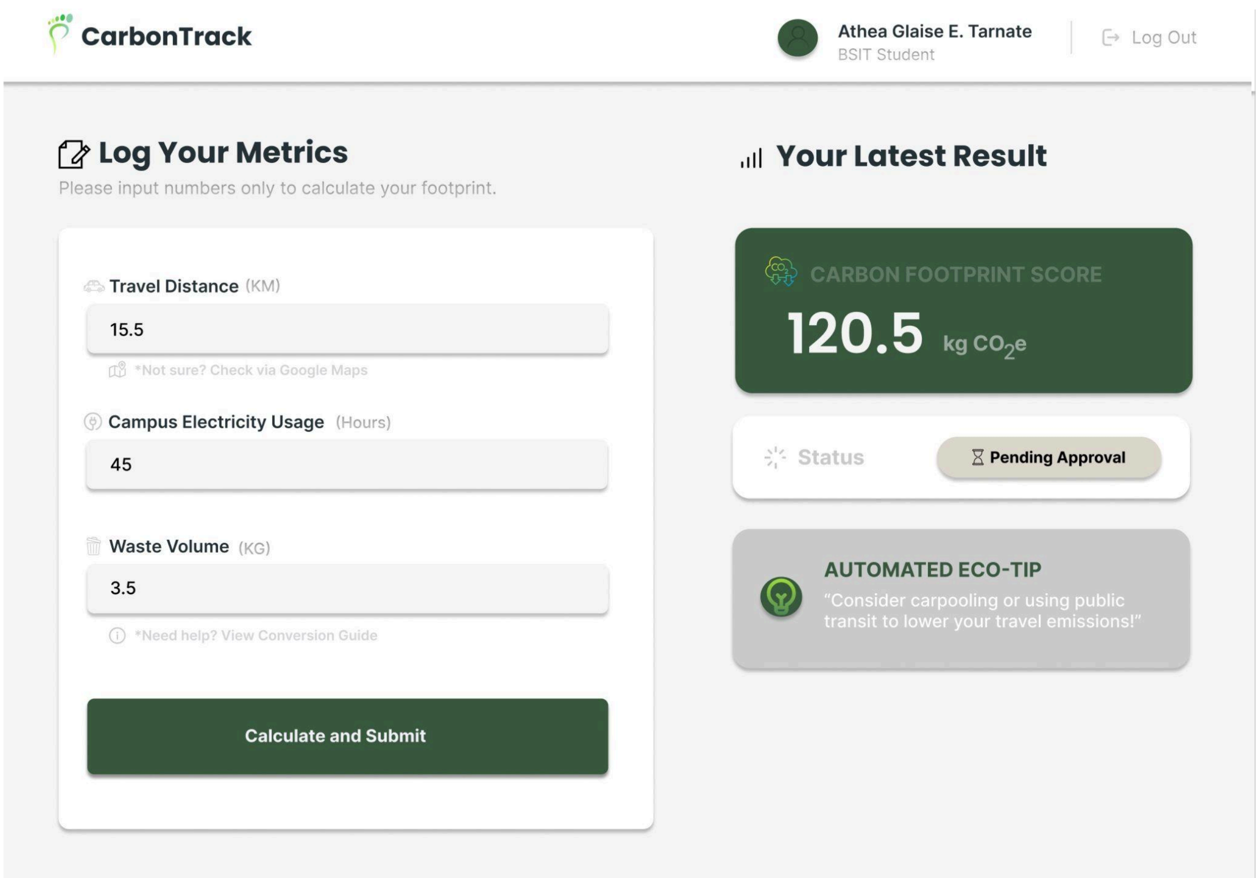


Module 2: The Data Entry Phase (Student Dashboard)

Purpose: Allows SOAST students to log their environmental metrics and view their personal carbon footprint score.

Key Features:

- **Metric Input Fields:** Provides specific numerical input fields for "Travel Distance (KM)," "Campus Electricity Usage (Hours)," and "Waste Volume (KG)".
- **Processing:** Users submit their data via the "Calculate and Submit" button.
- **Real-time Feedback:** Immediately displays the user's "CARBON FOOTPRINT SCORE" (e.g., 120.5) and tags the submission with a "Pending Approval" status.
- **Sustainability Engagement:** Generates an "AUTOMATED ECO-TIP" (e.g., "Consider carpooling or using public transit to lower your travel emissions!") to encourage environmentally friendly habits.

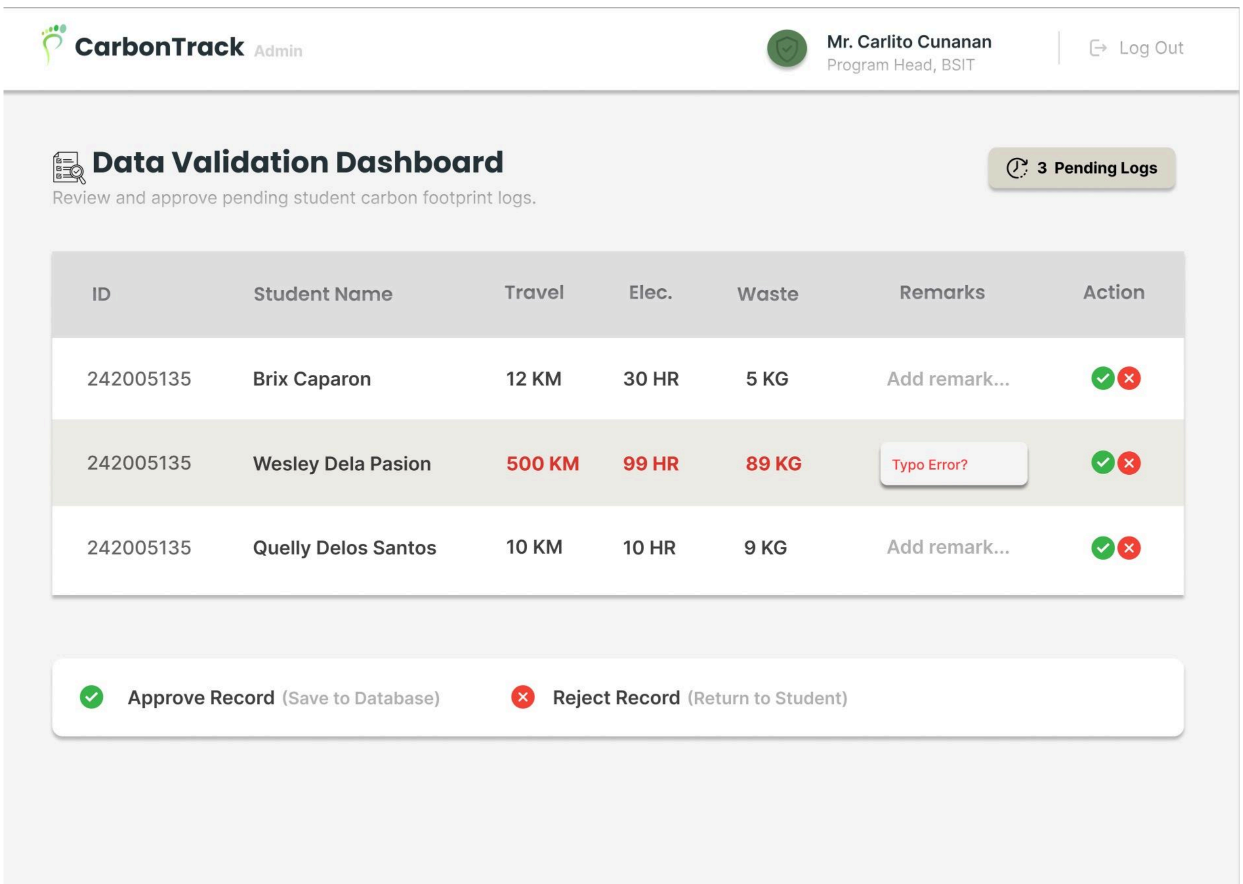


Module 3: The Validation Phase (Data Validation Dashboard)

Purpose: Enables the Program Head to review, verify, and manage the pending records submitted by the students.

Key Features:

- **Pending Queue:** Displays a counter of pending logs requiring action (e.g., "3 Pending Logs").
- **Data Grid:** Presents a comprehensive table detailing the Student ID, Student Name, Travel, Elec., and Waste metrics.
- **Action Controls:** The Program Head can leave "Remarks" on specific entries (e.g., flagging a "Typo Error?").
- **Workflow Execution:** The validator must execute a decision to either "Approve Record (Save to Database)" or "Reject Record (Return to Student)" for each entry.

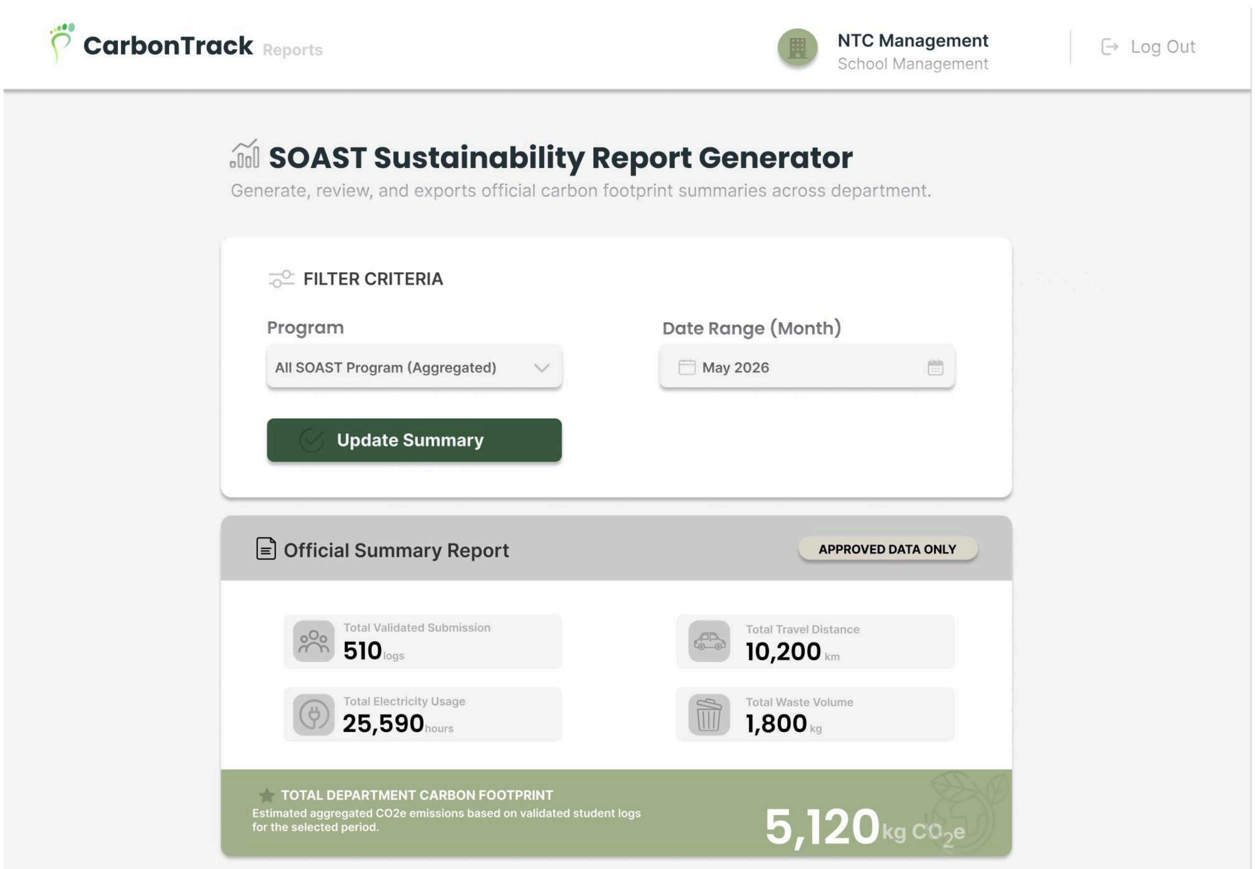


Module 4: The Aggregation Phase (Sustainability Report Generator)

Purpose: Utilized by the NTC Admin Office and School Management to generate, review, and export official carbon footprint summaries.

Key Features:

- **Filter Criteria:** Administrators can filter the data by "Program" (e.g., All SOAST Program) and "Date Range".
- **Data Integrity:** The Official Summary Report explicitly processes "APPROVED DATA ONLY," ensuring that rejected or pending logs do not skew the final metrics.
- **Summary Widgets:** Displays high-level aggregated data including "Total Validated Submission," "Total Electricity Usage," "Total Travel Distance," and "Total Waste Volume".
- **Executive Insight:** Highlights the "TOTAL DEPARTMENT CARBON FOOTPRINT," providing the estimated aggregated $CO_{2}e$ emissions for the selected period.



VI. Validation Plan

The Validation Plan ensures that the **Carbon Track: SOAST Students Carbon Footprint Monitoring System** meets all specified functional requirements and that every user journey is thoroughly tested. The Requirements Traceability Matrix (RTM) below links the core system requirements (derived from the UI/UX Blueprint) directly to specific Test Cases. This guarantees that no functional requirement is left untested before deployment.

Requirements Traceability Matrix (RTM)

Req ID	Requirement Description	Actor/ User	Test Case ID	Test Case Description	Expected Outcome
REQ-01	The system must authenticate users using NTC credentials and restrict access based on user roles (Student, Program Head, Management).	All Users	**TC-01**	Verify Role-Based Access Control (RBAC) during Login.	User is redirected to their respective dashboard successfully; unauthorized access is denied.
REQ-02	The system must require mandatory agreement to the NTC Data Privacy and Consent before allowing login.	All Users	**TC-02**	Validate the Data Privacy Checkbox enforcement.	Login is blocked if the checkbox is unchecked.
REQ-03	The system must allow students to input purely numerical values for Travel Distance (KM), Campus Electricity Usage (Hours), and Waste Volume (KG).	Student	**TC-03**	Validate numeric constraints on the input form.	System accepts numeric inputs and displays error for letters/special characters.

REQ-04	The system must calculate the user's Carbon Footprint Score upon clicking "Calculate and Submit" and assign a "Pending Approval" status.	Student	**TC-04**	Verify calculation accuracy and status assignment.	Accurate score is displayed (e.g., 120.5) and the log appears in the DB as "Pending".
REQ-05	The system must display a list of "Pending Logs" containing student metrics to the Program Head for review.	Program Head	**TC-05**	Verify data retrieval in the Validation Dashboard.	Table accurately reflects pending DB records including Student ID, Name, and metrics.
REQ-06	The system must allow validators to add "Remarks" and execute "Approve Record" or "Reject Record" actions.	Program Head	**TC-06**	Test Approve/Reject database commit logic.	Approved records are saved to the master database; Rejected records are returned/flagged.
REQ-07	The system must allow administrators to filter reports by "Program" and "Date Range (Month)".	Management	**TC-07**	Verify Report Filter logic.	Dashboard updates based on the selected Program and Date criteria.
REQ-08	The system must generate an "Official Summary Report" that aggregates "APPROVED DATA ONLY".	Management	**TC-08**	Verify data aggregation rules for the Management Dashboard.	Total Validated Submissions, Travel, Electricity, Waste, and Grand Total CO_2e match the approved database entries exclusively.

Detailed Test Case Definitions

Test Case ID: TC-08

Test Name: Validation of "Approved Data Only" Aggregation

Linked Requirement: REQ-08

Pre-conditions: * The database contains a mix of "Pending," "Rejected," and "Approved" carbon footprint logs.

The Management User is successfully logged in.
Test Steps:

1. Navigate to the "SOAST Sustainability Report Generator" module.
2. Set Filter Criteria: Select "All SOAST Program (Aggregated)" and Date Range "May 2026".
3. Click the "Update Summary" button.
4. Compare the generated summary cards (Total Travel, Total Electricity, Total Waste) against manual calculations of *only* the "Approved" records in the database.
Expected Result: The "TOTAL DEPARTMENT CARBON FOOTPRINT" and individual widgets must mathematically ignore any log that is still "Pending" or "Rejected". **Actual Result:** *(To be filled during actual testing phase)* **Status:** [Pass / Fail]